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—Dr Prahlad Vadakkepat, director, Robhatah Robotic Solutions and a faculty member of National University of Singapore



design, sensor data processing, code optimisation, communication, etc, are essential.

The mobile robots have to operate in dynamic environments, and intelligent and interactive modes of operations are crucial. Expertise in mechanical and electronic designs, sensor data processing/fusion, control, intelligent operation-related skill sets are important in this field. The operational environments and application domains of the mobile robots bring in challenges related to obstacle avoidance, real-time path planning and interacting with objects.

Vadakkepat says, "Presently, India has a pool of trained manpower in the robotics segment. So, proper training along with professional degree would be a turnkey for a fresher. The training should be totally industry focused. I feel an effective collaboration between industry and academia may be fruitful for both. The existing industry-institute partnership models of some reputed foreign universities are ideal to follow." Let's collate their suggestions to dig the field in a comprehensive manner.

Training and workshop

The training should be balanced enough to deal with both theoretical and practical aspects of robotics. Being an inter-disciplinary course, students from electronics engineering background opting for a career in robotics need to have a sound knowledge of mechanical, electrical, instrumentation and computer engineering. As

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"Taking up college projects in robotics will help students to determine their true motivation and prepare them early. A number of technical festivals across India with robotics as the central theme provide avenues for aspirants to develop their skill sets. There are players extending embedded systems and robotics-related training. Internet is a good source for various tools and technical papers. Attending technical meetings/conferences and seminars will assist in expanding the domain knowledge."

Keep in mind that robotics is a completely new package to improve services. That is why at times you are required to use all your skills to come up with a solution that is trivial and requires an insight into various concepts. Introduction to the mechanical parts of robots is a must.

"If someone wants to be a good robot developer then he has to possess a good understanding of various mechanical aspects and fabrication techniques,

which involve working with sheet metals, die castings, CNC lathes, milling machines and moulding engineering," says Gaur. Even exposure to soldering on PCBs may be useful.

Try to get some idea about different types of actuators used in robotics, with focus on servo motor and stepper motor. Moreover, working knowledge of different sensors used in robotics may provide an extra advantage. Especially, detailed knowhow of IR sensors, proximity sensors, LDR, temperature sensors, motion detectors and acceleration detectors is quite pertinent.

To control your robot, you need to know about microcontrollers. Try to put emphasis on the interfacing action of microcontrollers with sensors and actuators using programming languages like MPLAB.

Another suggestion is that, irrespective of the stream of engineering you are pursuing, be bold to explore multidisciplinary areas—mechanical design, electronics, microcontroller programming, wireless control and sensors. The first and foremost step is to learn how to control a motor, may be directly from a PC. Microcontroller programming comes next. Identify peers with similar interests, and form a team for group learning and sharing the workload. Pass on the expertise to your juniors. Assembly level to higher level programming, soldering and mechanical design skills should be picked up. Internet is a powerful medium to keep abreast with the latest in robotics and idea exchange.

From classroom to industry

Perhaps, your awareness of robotics and artificial intelligence are from a notional perspective only. This is a

Fresh electronics graduates can definitely play important roles in some of the positions, especially in the creative designing and problem-solving sections.

—Pulak Gaur, CEO and co-founder of Gridbots

